# AMC 10 Style Practice Test

25 problems · 75 minutes · no calculator · choose one answer (A–E) for each problem

Name: ________________ Date: ________________

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**1.** What is the value of $\frac{2+4+6+\ldots+20}{1+3+5+\ldots+19}$?

**(A)** $\frac{9}{10}$ **(B)** 1 **(C)** $\frac{11}{10}$ **(D)** $\frac{6}{5}$ **(E)** $\frac{11}{9}$

**2.** A price first increased by 20%, and then the new price decreased by 25%. How does the final price compare to the original?

**(A)** 10% lower **(B)** 5% lower **(C)** unchanged **(D)** 5% higher **(E)** 10% higher

**3.** Anna walked the first 3 km at 6 km/h and the next 3 km at 2 km/h. What was her average speed for the whole walk?

**(A)** 2.5 km/h **(B)** 3 km/h **(C)** 3.5 km/h **(D)** 4 km/h **(E)** 4.5 km/h

**4.** A rectangle has perimeter 34 and area 60. What is the length of its diagonal?

**(A)** 10 **(B)** 11 **(C)** 12 **(D)** 13 **(E)** 14

**5.** How many two-digit numbers are divisible by 7 but not divisible by 3?

**(A)** 7 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 13

**6.** The average of five numbers is 12. One number is removed, and the average of the remaining four becomes 13. What number was removed?

**(A)** 5 **(B)** 6 **(C)** 7 **(D)** 8 **(E)** 10

**7.** How many digits does the number $2^{10} \cdot 5^7$ have?

**(A)** 8 **(B)** 9 **(C)** 10 **(D)** 11 **(E)** 17

**8.** A class has 8 students. A team of 3 must be chosen, but two of the students have quarreled and cannot both be on the team. In how many ways can the team be chosen?

**(A)** 40 **(B)** 44 **(C)** 48 **(D)** 50 **(E)** 56

**9.** A circle is inscribed in a square with side 8. What is the area of the part of the square lying outside the circle?

**(A)** $64 - 8\pi$ **(B)** $64 - 16\pi$ **(C)** $64 - 32\pi$ **(D)** $32 - 16\pi$ **(E)** $16 - 4\pi$

**10.** In an arithmetic sequence the first term is 5 and the common difference is 3. Which term of the sequence is the number 302?

**(A)** 99th **(B)** 100th **(C)** 101st **(D)** 102nd **(E)** 103rd

**11.** A real number $x$ satisfies $x + \frac{1}{x} = 5$. What is $x^3 + \frac{1}{x^3}$?

**(A)** 110 **(B)** 115 **(C)** 120 **(D)** 125 **(E)** 130

**12.** Two standard dice are rolled. What is the probability that the sum of the numbers shown is a prime number?

**(A)** $\frac{1}{3}$ **(B)** $\frac{2}{5}$ **(C)** $\frac{5}{12}$ **(D)** $\frac{1}{2}$ **(E)** $\frac{7}{12}$

**13.** A right triangle has legs 6 and 8. What is the radius of its inscribed circle?

**(A)** 1.5 **(B)** 2 **(C)** 2.4 **(D)** 2.5 **(E)** 3

**14.** What is the least positive integer $n$ such that $n!$ is divisible by 2025?

**(A)** 10 **(B)** 11 **(C)** 12 **(D)** 15 **(E)** 25

**15.** How many integers $n$ with $1 \le n \le 100$ are such that $n^2 + n$ is divisible by 6?

**(A)** 33 **(B)** 50 **(C)** 66 **(D)** 67 **(E)** 83

**16.** What is the area of a regular hexagon with side 4?

**(A)** $16\sqrt{3}$ **(B)** $24\sqrt{3}$ **(C)** $32\sqrt{3}$ **(D)** $48\sqrt{3}$ **(E)** 96

**17.** How many four-digit even numbers have all digits distinct and taken from the set $\{1, 2, 3, 4, 5\}$?

**(A)** 24 **(B)** 36 **(C)** 48 **(D)** 60 **(E)** 72

**18.** What is the sum of all positive divisors of 360, including 1 and 360 itself?

**(A)** 1024 **(B)** 1080 **(C)** 1120 **(D)** 1170 **(E)** 1260

**19.** What is the sum of all real roots of the equation $\left|\,|x - 2| - 3\,\right| = 1$?

**(A)** 0 **(B)** 4 **(C)** 6 **(D)** 8 **(E)** 10

**20.** In triangle $ABC$, point $D$ lies on side $BC$ so that $BD : DC = 1 : 2$. Point $E$ is the midpoint of $AD$. Line $BE$ meets side $AC$ at point $F$. What is the ratio $AF : FC$?

**(A)** 1 : 2 **(B)** 2 : 5 **(C)** 2 : 3 **(D)** 1 : 4 **(E)** 1 : 3

**21.** A sequence is defined by $a_1 = 1$ and $a_{n+1} = 2a_n + 1$ for all $n \ge 1$. What is $a_{10}$?

**(A)** 511 **(B)** 512 **(C)** 1022 **(D)** 1023 **(E)** 2047

**22.** How many integers $x$ satisfy the inequality $|x - 3| + |x + 2| < 9$?

**(A)** 7 **(B)** 8 **(C)** 9 **(D)** 10 **(E)** 11

**23.** What are the last two digits of $3^{2025}$?

**(A)** 01 **(B)** 07 **(C)** 27 **(D)** 43 **(E)** 83

**24.** Lattice paths from (0, 0) to (5, 5) consist of unit steps right and up. How many such paths never go above the line $y = x$?

**(A)** 14 **(B)** 32 **(C)** 42 **(D)** 120 **(E)** 252

**25.** What is the sum of all real numbers $x$ for which $(x^2 - 5x + 5)^{x^2 - 9x + 20} = 1$?

**(A)** 9 **(B)** 10 **(C)** 13 **(D)** 14 **(E)** 15

## Answer Key

*Parent page: cut off or do not print for the student. Full solutions are available on the test page online.*

| | | | | |
|---|---|---|---|---|
| 1 | 2 | 3 | 4 | 5 |
| **C** | **A** | **B** | **D** | **C** |
| 6 | 7 | 8 | 9 | 10 |
| **D** | **A** | **D** | **B** | **B** |
| 11 | 12 | 13 | 14 | 15 |
| **A** | **C** | **B** | **A** | **C** |
| 16 | 17 | 18 | 19 | 20 |
| **B** | **C** | **D** | **D** | **E** |
| 21 | 22 | 23 | 24 | 25 |
| **D** | **B** | **D** | **C** | **E** |